

1. for Loop – When Iteration Count is Known

When Interviewers Expect You to Use for

Use a for loop when:

- Number of iterations is known beforehand.
- Traversing arrays, strings, collections (using index).
- Counting problems.

Example:

```
for (int i = 0; i < 10; i++) {  
    System.out.println(i);  
}
```

Hidden Facts Interviewers Love

Q1. Are all three parts of a for loop mandatory?

No.

```
for (;;) {  
    System.out.println("Infinite");  
}
```

Equivalent to

```
while (true) {  
}
```

Q2. Can multiple variables be declared?

Yes.

```
for (int i = 0, j = 10; i < j; i++, j--) {  
    System.out.println(i + " " + j);  
}
```

Output

```
0 10  
1 9  
2 8  
3 7  
4 6
```

Q3. Can initialization contain method calls?

Yes.

```
for (initialize(); condition(); update()) {  
}
```

All three sections can contain method calls.

Q4. Which executes first?

Execution order:

```
Initialization  
↓  
Condition  
  
↓  
Body  
  
↓
```

Update

↓

Condition

↓

Body

Interview favorite.

Q5. What is the scope of i?

```
for (int i = 0; i < 5; i++) {  
  
}  
  
System.out.println(i);
```

Compiler Error.

i exists only inside the loop.

Q6. Can update be omitted?

Yes.

```
int i = 0;  
  
for (; i < 5;) {  
    i++;  
}
```

Q7. Can update happen inside body?

Yes.

```
for (int i = 0; i < 5;) {  
    System.out.println(i);  
    i++;  
}
```

Tricky Output Questions

Question

```
for(int i=0;i<5;i++);  
{  
    System.out.println("Hello");  
}
```

Output

```
Hello
```

Why?

Because of the semicolon.

The loop body is empty.

Question

```
for(int i=1;i<=5;i*=2)  
    System.out.print(i+" ");
```

Output

```
1 2 4
```

Question

```
for(int i=5;i>0;i--)  
    System.out.print(i-- + " ");
```

Output

```
5 3 1
```

Reason:

i-- inside print decreases it once.

Loop update decreases it again.

Hard Interview Questions

Why is for generally preferred over while when count is known?

Answer:

- Better readability
 - Initialization, condition, update together
 - Easier maintenance
 - Less chance of forgetting update
-

Can a for loop become infinite?

Yes.

```
for(;;)
```

or

```
for(int i=0;i>=0;i++)
```

Eventually integer overflow makes *i* negative after billions of iterations.

2. while Loop – When Iteration Depends on a Condition

Interview Rule

Use while when the number of iterations is **unknown**.

Examples

- Reading a file
 - User input
 - Waiting for connection
 - Polling
-

Hidden Facts

Q1. What happens if condition is false initially?

```
while(false)
{
}
```

Runs **zero times**.

Q2. Which part is missing compared to for?

Initialization and update.

Programmer handles them manually.

Q3. Most common mistake?

Infinite loop.

```
int i=0;

while(i<5)
{
    System.out.println(i);
}
```

Forgot i++.

Q4. Is this valid?

```
while(true);
```

Yes.

Infinite empty loop.

Q5. Can condition contain assignment?

```
while((line = reader.readLine()) != null)
```

Very common in Java IO.

Interview favorite.

Tricky Output

Question

```
int i=1;

while(i<=5)
{
```

```
        System.out.print(i++ + " ");  
    }
```

Output

```
1 2 3 4 5
```

Question

```
int i=5;  
  
while(i-->0)  
{  
    System.out.print(i+ " ");  
}
```

Output

```
4 3 2 1 0
```

Very famous interview question.

Question

```
int i=0;  
  
while(i++<5)  
{  
    System.out.print(i+ " ");  
}
```

Output

1 2 3 4 5

Need to understand post-increment.

Hard Interview Questions

Why is while preferred for reading files?

Because the number of lines is unknown.

Can every for loop be converted to a while loop?

Yes.

Example

```
for(int i=0;i<5;i++)
```

Equivalent

```
int i=0;

while(i<5)
{
    i++;
}
```

3. do-while Loop – Menu Driven Programs

Interview Rule

Use do-while when **the loop must execute at least once**.

Examples

- ATM
 - Calculator
 - Login retries
 - Restaurant menu
 - Game menu
-

Hidden Facts

Q1. Why is it perfect for menu-driven programs?

Because the menu should appear **at least once** before asking whether to continue.

Q2. Condition checked before or after?

After.

This is the biggest difference.

Q3. Minimum executions?

Loop	Minimum Iterations
for	0
while	0
do-while	1

Very common interview question.

Example

```
int choice;

do
{
    System.out.println("1. Add");
    System.out.println("2. Exit");
```

```
choice = scanner.nextInt();  
  
}while(choice!=2);
```

Tricky Output

Question

```
int i=5;  
  
do  
{  
    System.out.print(i+" ");  
}while(i<5);
```

Output

```
5
```

Condition is false.

Still executes once.

Question

```
int i=1;  
  
do  
{  
    System.out.print(i++ +" ");  
}  
while(i<=5);
```

Output

1 2 3 4 5

Question

```
int i=5;

do
{
    i--;
}
while(i>5);

System.out.println(i);
```

Output

4

Comparison (Very Common Interview)

Feature	for	while	do-while
Condition Checked	Before	Before	After
Executes At Least Once	✗	✗	✓
Best For	Known iterations	Unknown iterations	Menu-driven programs
Initialization	Inside loop	Usually outside	Usually outside
Update	Inside loop	Manual	Manual
Infinite Loop	for(;;)	while(true)	do{}while(true);

Interview Trap Questions

Which loop is fastest?

None.

The JVM optimizes them similarly. Performance differences are negligible; choose based on readability and the problem.

Which loop should you use in production?

- Known count → for
- Unknown count → while
- Must execute at least once → do-while

Choosing the loop based on intent makes the code easier to understand.

Can every for loop become a while loop?

Yes.

Can every while loop become a for loop?

Yes.

A for loop is more flexible than many developers realize.

Can every do-while loop become a while loop?

Yes, but you may need to duplicate the first execution or restructure the logic because while checks the condition before the first iteration.

Top 15 Loop Interview Questions

1. Why is for preferred when the iteration count is known?
2. Can all three sections of a for loop be omitted?
3. What is the scope of the loop variable declared in a for loop?
4. Can you declare multiple variables in a for loop?
5. What is the execution order of a for loop?
6. Why is while preferred for file reading or user input?

7. What is the most common cause of an infinite while loop?
8. Explain while(i-->0) and predict its output.
9. What is the minimum number of iterations for each loop type?
10. Why is do-while suitable for menu-driven programs?
11. Can a do-while execute if its condition is false?
12. Can all loop types be converted into each other?
13. What is the difference between for(;;) and while(true)?
14. What happens if a semicolon (;) is accidentally placed after a loop declaration?
15. Which loop would you choose for traversing an array, processing unknown-length input, and displaying a menu, respectively?